

Ahmed Ikram | Data Engineering & Analytics

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Specialises in event-driven pipeline engineering, ETL/ELT orchestration, and cloud-native data modelling - having built and deployed a production-grade Kafka ingestion system processing 2.5M+ events at 100+ msg/sec with fault-tolerant schema design, dead-letter routing, and exactly-once semantics on AWS. Proven ability to engineer data systems that handle schema drift, concurrency failures, and real-world data quality issues end-to-end. Seeking a Data Engineering or Analytics post-graduation role to apply pipeline design, PostgreSQL data modelling, and Airflow orchestration skills in a production environment.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, Apache Kafka, PostgreSQL, FastAPI, Redis, MLflow, Docker, AWS | Mar 2026 - May 2026

- Led backend and data engineering on a 7-person Agile team for NCR Atleos; independently replaced the direct-DB generator with a full event-driven pipeline - designing the Kafka message bus, data lake schema, ML infrastructure and test suite entirely post-submission improving upon the team-delivered project
- Architected an Apache Kafka KRaft ingestion pipeline processing 2.5M+ events every 24 hours at 100+ msg/sec across 7 log sources; 7 source-specific parsers with .get() safe defaults ensure schema drift never halts ingestion; malformed records route to a dead-letter table; hybrid Redis SET + 10K LRU deduplication guarantees exactly-once semantics across consumer restarts with exponential backoff retry on all failures
- Designed a PostgreSQL 16 data lake with JSONB payloads (new log sources require only a new parser - zero schema migrations), a v_unified_analysis UNION ALL view as the single ML query target, and 14 composite B-tree indexes sustaining performance at 1M+ rows; validated pipeline correctness via 670 tests across 10 tiers, catching silent data loss under 50 concurrent writes and validating exactly-once semantics end-to-end

W3C Web Logs ETL Pipeline | Apache Airflow, Python, PostgreSQL, Power BI, Power Automate, AWS RDS, DAX | Feb 2026 - June 2026

- Designed an Airflow DAG orchestrating incremental ETL of over 90 raw web logs totalling 156,000 log records into a 9-dimension Star Schema on AWS RDS PostgreSQL - nine dimension tasks fan out simultaneously post-staging, with the fact table load fanning back in once all nine complete, making phase three 8x faster
- Engineered ingestion handling two distinct IIS column formats (14 and 18 columns) detected dynamically from hash fields headers; incremental filename deduplication makes reruns fully safe
- Resolved data quality issues during transformation: IP-to-geolocation enrichment via ip-api.com batch API (100 IPs/request) with rate-limit-aware 1.5s pausing and AWS RDS reconnection with 3-retry logic after idle drops during long API phases; left joins with -1 surrogate key fallback throughout ensure zero records are silently dropped from the fact table
- Built Power BI reports across 7 pages surfacing traffic trends, P95 response times via DAX, geographic distribution across 78 countries, and bot vs. human traffic split (38% crawler); automated dataset refresh via Power Automate with a failure-handling flow sending success/failure email notifications

Haggis Species Classification & Predictive Modelling | Python, scikit-learn, XGBoost, pandas, NumPy, matplotlib, seaborn, Jupyter Notebook | Nov 2025 - Dec 2025

- Ran a complete ML pipeline from EDA through feature engineering to model evaluation; engineered and validated 2 novel ratio-based features using sklearn pipelines (ColumnTransformer, RobustScaler) that ranked among the top-3 predictors across all models, whilst preventing data leakage
- Trained and compared 7 classifiers (XGBoost, Random Forest, Logistic Regression and others) achieving ~90% accuracy and tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering and built a Linear Regression model ($R^2=0.756$) and validated assumptions via residual diagnostics and standardised coefficient analysis

PROJECTS

- Project Management Platform with GitHub Integration** | Python, Flask, PostgreSQL, React, Socket.io, AWS, Docker, GitHub Actions, CI/CD | Jan 2025 - Apr 2025
- Designed a 12-table PostgreSQL schema built around actual API access patterns - indexes on FK, join, and filter columns throughout; dedicated tables for audit logs (filterable by user, action, resource, and date with configurable retention and cleanup), typed saved reports (JSON summary and details fields), system settings, GitHub repository tracking, and bidirectional task-to-Issue/PR link associations enforced at the database layer
 - Deployed production AWS data infrastructure: ECS Fargate & ALB serving the Flask API, RDS PostgreSQL in a private subnet accessible only from ECS at the network layer, and a React SPA on S3 & CloudFront; rolling deploys tag each image with the commit SHA and latest for immutable rollback references at any prior build
 - 1,446 automated tests at 85%/85% line coverage hard-gate every PR via GitHub Actions before any AWS step runs; OIDC federation authenticates the pipeline to AWS without static credentials stored anywhere

- Global Superstore Business Intelligence & KPI Interactive Dashboard** | Excel, Power Query, Pivot Tables | Oct 2025
- Analysed 51,000 sales records across regions, categories, and time periods; identified that aggressive discount strategies were directly suppressing net profit in the Furniture segment
 - Built an interactive slicer-driven dashboard enabling non-technical stakeholders to independently explore the data without requiring analytical support; presented findings as actionable business recommendations

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Bash, Typescript
Data Engineering	Apache Airflow, ETL/ELT, Pipeline Design, Apache Kafka, Dead-Letter Queues, Stream Processing, Pipeline Design
Data Modelling	Star Schema, Data Lake Design, JSONB Payloads, Dimensional Modelling, Schema-on-Read
Databases	PostgreSQL, MongoDB, MySQL, SQLite, NoSQL, Redis
Analytics & BI	Pandas, NumPy, Matplotlib, seaborn, Power BI, Power Automate, DAX, Jupyter Notebook
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, IAM, SageMaker), Docker, GitHub Actions, CI/CD, Linux, Azure

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations	Feb 2025
AWS Academy Graduate - Cloud Foundations	Mar 2024
Foundations of Azure AI: Concepts, Capabilities, and Implementation	Oct 2024

EXPERIENCE

- Beaully Mini Market**, Dundee, Scotland
Retail Assistant
September 2023 - Present
- Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption
- Zulus Peri Peri Grill**, Dundee, Scotland
Front of House Worker
June 2021 - September 2023
- Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure

TRANSFERRABLE SKILLS

Pipeline Design & Data Modelling Design normalised schemas and ingestion architectures with reliability, schema drift tolerance, and edge cases as first-class concerns	Data Quality & Validation Apply multi-stage validation - row count assertions, null checks, surrogate key fallbacks - to ensure no records are silently dropped or corrupted downstream
Root Cause Analysis Systematically isolate pipeline failures and data issues; experienced diagnosing silent data loss, concurrency bugs, and API rate-limit failures under time pressure	Stakeholder Communication Translate pipeline outputs and analytical findings into plain business language for non-technical audiences and project stakeholders